

UNIVERSITY OF MALAYA

WID3014 PRACTICAL AI
WEEK 3 CLASS ACTIVITY

Name	Matrics Number
Adam Bin Razali	22002146
Nazifah Binti Mohamed Murathu Khan	23004948
Husna Nadhirah Binti Samaon	22002695
Nur Nabilah Binti Noor Ramli	22002697
Muhammad Aqlan Bin Azizul	23002551

How to build a system to show users trending hashtags?

1) Business Objectives & ML Objectives

Business Objectives:

a. Surface real-time cultural moments, breaking news, and viral events to drive session duration, Daily Active Users, and click-through rates on the "Explore" or "Search" surfaces.
b. Provide high-intent surfaces for sponsored trends, contextual advertising, and partner campaigns.
c. Maximize the speed of detection so trends are surfaced while they are happening, not hours after interest has peaked.

Machine Learning Objectives:

a. Spike and Velocity Detection:
- Formulate trend detection not as raw count aggregation, but as acceleration or relative burst compared to a baseline.
- Formulate as a time-series anomaly detection task or Poisson rate change problem to detect statistically significant surges.

b. Candidate Ranking:
- Frame trend selection as a ranking problem using a composite scoring functions
- Optimize ranking models for user engagement, such as predicting P(Click | User, Trend)

c. Semantic Deduplication and Clustering (NLP):
- Cluster semantically identical, synonymous, or misspelled hashtags into a single canonical trend topic.

d. Inauthentic Behavior and Spam Filtering (Classification):
- Binary or multi-class classification to predict the probability that a sudden surge is driven by bots, automated scripts, or coordinated manipulation rather than organic user interest.
- Score content diversity (e.g., ratio of unique authors to total posts) to penalize volume dominated by single power-users.

2) How to measure your model's performance? Do we have ground truths?

Measuring our trending hashtag system is tricky because there is no objective, perfect definition of a "trend." Since we cannot rely on a standard answer key, we have to build proxy ground truths and measure our spike detection, ranking, clustering, and spam filtering individually.

Defining Our Ground Truths

- **Statistical Proxy:** We define a true trend mathematically, looking for hashtags that suddenly spike by over 500% compared to their normal weekly average.
- **Human Review:** Human raters manually review historical data to tell us if a spike was organic, bot-driven, or just an everyday popular tag.
- **User Feedback:** We look at historical log data, letting actual user clicks and downstream content creations act as the ultimate judge.

Offline Metrics (Pre-Launch Testing)

- **Detection Speed:** We track the exact minutes it takes the model to flag a spike after the mathematical surge begins.
- **Ranking Quality:** We use metrics like NDCG or MRR to check if the most clickable, high-intent trends actually make it to the top of the list.
- **Precision and Recall:** We measure how accurately the spam filter catches bots without hiding real trends, and how well our clustering groups similar topics.

Online Metrics (Live Production)

- **Engagement:** We monitor Click-Through Rates on the Explore page and track overall user session duration.
- **Freshness:** We watch the list's turnover rate to ensure it actively surfaces breaking news instead of staying frozen for hours.
- **Author Diversity:** We check the ratio of unique creators to total posts so trends aren't artificially inflated by a few highly active power-users.

3) Constraints

a) Infrastructure and Performance

- **Real-Time Processing Latency:** The system must ingest and analyze millions of concurrent posts with sub-minute latency. Complex ML architectures (like massive transformer models for semantic clustering) are often too slow for the live data firehose, forcing a strict trade-off between model accuracy and processing speed.
- **Compute and Storage Costs:** Running continuous time-series anomaly detection and NLP clustering across a global stream requires massive distributed processing (e.g., Apache Kafka, Flink). The compute budget directly constrains how deep or personalized the ranking functions can be in real-time.

b) Algorithmic and Data Realities

- **Adversarial Adaptation (The Cat-and-Mouse Game):** Bad actors, bot nets, and marketing farms constantly evolve their tactics to bypass spam filters and hijack trends. Static classification models degrade quickly, requiring continuous feature engineering and retraining.
- **Freshness vs. Stability Trade-off:** The time-decay parameter is highly sensitive. If historical data decays too fast, the trending list flickers wildly; if it decays too slowly, the list feels stale and misses breaking news.
- **The "Evergreen" Problem:** Terms like #MondayMotivation or #Kpop has a massive daily volume. The system is constrained by the need to dynamically establish baselines so these everyday popular tags do not permanently block emerging, smaller-volume cultural moments.

c) Safety and Business Alignment

- **Brand Safety and Content Moderation:** The system is strictly constrained by trust and safety policies. It must actively suppress NSFW content, hate speech, and dangerous misinformation from surfacing on the Explore page, even if the engagement spike is 100% organic and rapid.

4) What is the simplest possible model?

Few notes -

1) Count Based
    a) It is an algorithm that tracks the amount of time a keyword is being used, like a count. It is determined by the number of frequencies.
    b) The cons of it is that it doesn't care of the sentiment of the post, this algorithm can be highly negatively affected by bots.
2) Rule-Based
    a) Include ruling based during the inference engine
    b) Lets say if the hashtag has a high count used, but it contains banned keywords, then we won't identify it as trending hashtag

Answer :

The answer would be a count based model. This is the easiest model, as it uses count to determine the relevancy of the hashtags, without any rule based inference. We can use a sliding time window based model, where we count the total number of hashtags used in a timeframe.

5) Localization/personalization of trending hashtags?

Localisation and personalisation of trending hashtags are both needed to balance macro-filtering of cultural awareness with macro-filtering of individual relevance, which can prevent both regional noise and irrelevant information noise for users or personal echo chambers (eg. users miss critical breaking news).

a) Localization of trending hashtags

Partition the stream processing pipelines by filtering the trending hashtags to geographic boundaries, language, or cultural grouping. As an example, a user from Kuala Lumpur can fetch global trends and local Malaysian trending partitions to ensure regional discussions take precedence over distant global noise.

b) Personalization of trending hashtags

The personalisation layer functions as a recommendation pipeline that transforms a rigid or regional top-10 list into a tailored feed for individual users. The layer consists of four components:

- Candidate retrieval: Retrieve a precomputed pool of top trending candidates according to the regions to narrow the search space.
- Feature engineering: Extract real-time and batch features, including user interest vectors and metadata of hashtags.
- Scoring and Re-ranking Model: Predict the likelihood of user engagement towards specific hashtags using a lightweight machine learning model such as Gradient Boosted Decision Tree.
- Hybrid Blending and Diversity Controls: Combines the trend score of the region with the individual user engagement score using a weighted formula to prevent echo chambers.